Reciprocal Differential Energy on Finite Double Covers

Boundary reductions, quadratic models, and the theta heat flow

Drafted for Edward Baker

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Abstract

For a holomorphic function on a bounded spatial domain, the differential $dz/\sqrt{H}$ on its two-sheeted cover has energy $2\int |H|^{-1}\,dA$. This energy is finite at a simple zero and diverges at a multiple zero. The research objective is an independent quantitative upper estimate for the canonical theta heat flow, rather than another formulation of simplicity. We develop the finite-domain analytic framework for this objective. On a two-branch annulus, a single-valued real primitive gives an exact reduction to a projected relative period and a complete exterior boundary correction. Subtracting a quadratic differential with the same core period yields explicit energy, relative-period, and full-boundary remainders bounded independently of the branch separation. A complete moment-controlled chart at a hypothetical nondegenerate heat event supplies all constants and a finite upper bound; a sharper bound strictly improves the initial coarse certificate on an explicit window. An exact quadratic calculation shows why a bounded core period and bounded exterior norm budgets do not bound the full energy. A previously recorded interval computation on one regular theta fiber supports the finite acquisition framework. The analytic estimates remain compatible with ordinary double collisions: no endpoint-uniform theta estimate, RH theorem, or simplicity exclusion is obtained.

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1 Project goals and the results established so far

The project studies the reciprocal differential on finite covers of the canonical theta heat field. The intended long-term application is to control degeneration as zeros approach one another. The distinction between an identity and an estimate is central. Expressing simplicity as local integrability does not prove simplicity. A useful advance would be an upper estimate obtained from the unchanged canonical source and remaining controlled in the relevant limiting regime, or a demonstrable finite acquisition advantage over direct integration using the same information.

This manuscript develops the analytic part of that program. Its setting is a bounded disk containing two simple real zeros. The resulting cover is an annulus, with two physical boundary components, one absolute core cycle, and a relative path connecting the two boundary components. The relative period must be retained: it can diverge even when the core period is constant and the exterior differential has uniformly bounded traces.

1.1 Canonical normalization and the motivating question

We use

$$H(t,z)=\int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du, \tag{1.1}$$

$$\Phi(u)=\sum_{n=1}^{\infty}(2\pi^2n^4e^{9u}-3\pi n^2e^{5u})e^{-\pi n^2e^{4u}}. \tag{1.2}$$

The classical theta identity fixes the normalization

$$H(0,z)=\frac{1}{8}\xi\left(\frac{1}{2}+\frac{\mathrm{i}z}{2}\right),\qquad \xi(s)=\frac{s(s-1)}{2}\pi^{-s/2}\Gamma(s/2)\zeta(s). \tag{1.3}$$

Appendix A derives this identity and the complete source reflection $\Phi(u)=\Phi(-u)$ from the Gaussian theta identity. The convention agrees with [1, (1)–(4)]. Our derivative and energy estimates below start from (1.1)–(1.2). In particular,

$$\partial_tH=-\partial_z^2H. \tag{1.4}$$

The RH motivation concerns the zero set of (1.3); none of the local theorems below assumes RH. In a hypothetical ordinary collision, the zeros split at the square-root scale in positive physical time. The reciprocal energy then grows logarithmically. Preventing such a collision would require a source estimate incompatible with that growth, with constants valid in the limiting family. The present finite estimates do not have that property.

1.2 Analytic contributions

The results are organized as follows.

1. Section 2 constructs the cover, normalizes the energy, and proves the local integrability statement at all multiplicities. It gives explicit coordinates for the full bordered annulus.
2. Section 3 proves the complete cut-boundary identity and a real-primitive reduction. With consistently oriented periods q,p,

$$\mathcal{E}=\mathrm{Im}(\bar{q}p)+R_{\mathrm{ext}},\qquad |R_{\mathrm{ext}}|\le 2\pi^2K_\partial^2. \tag{1.5}$$

   The bound uses the exterior differential trace; the relative period contains the missing degeneration.
3. Section 4 evaluates the entire quadratic cover and each of its cut and physical-boundary contributions. It proves a scoped failure of upper estimates based only on fixed exterior norm budgets and the permitted core period.
4. Section 5 subtracts a quadratic differential with the same core period. It proves explicit bounds for the full energy, relative period, and both boundary contributions, with constants independent of the root separation when the unit and its derivatives are controlled.
5. Section 6 derives a complete quantitative chart from canonical source moments at a hypothetical ordinary event. It proves a finite logarithmic upper bound and a strict improvement over its coarse version on a specified smaller window. These are shared direct estimates, not a new restriction on the source's admissible events.

Section 7 records analytic error consequences. Numerical support is confined to Section 8, with the detailed computational exposition left to its cited record. Section 9 identifies the remaining endpoint and coverage problems.

1.3 Comparison principle and scope

The direct comparator is always the same quantity, $2\int_\Omega |H|^{-1}\,dA$, supplied with all the same source, normalization, root, boundary, and period observations. It may use the same coordinate substitutions and exact identities. Consequently, a better bound than one coarse direct certificate does not establish a general advantage over direct integration. Likewise, a native numerical implementation can be faster without having more information.

The results collected here consolidate [4]; their proofs are included in full. Classical complex analysis, the exact canonical normalization above, and ordinary calculus supply the background. There is no compact finite-genus model of the complete entire theta cover in the argument, and no omission of the spatial boundary of a finite cover.

2 Finite double covers and reciprocal energy

2.1 Regular fibers and the factor of two

Let $\Omega \subset \mathbb{C}$ be a bounded domain with smooth boundary, and let H be holomorphic on a neighborhood of $\overline{\Omega}$, nonzero on $\partial\Omega$. When every zero in Ω is simple, set

$$X = \{(z,w) : z \in \overline{\Omega},\ w^2 = H(z)\}, \qquad \omega = \frac{dz}{w}. \tag{2.1}$$

The complex structure on X gives its orientation. At a zero α, write $H(z) = (z-\alpha)U(z)$, $U(\alpha) \neq 0$. In the local coordinate $z = \alpha + \zeta^2$,

$$w = \zeta\sqrt{U(\alpha+\zeta^2)}, \qquad \omega = \frac{2\,d\zeta}{\sqrt{U(\alpha+\zeta^2)}}. \tag{2.2}$$

Thus the branch point is regular on the cover and ω is holomorphic.

Proposition 2.1 (Full energy)**.** *The conformally invariant energy of ω is*

$$\mathcal{E}(H;\Omega) := \frac{\mathrm{i}}{2}\int_X \omega\wedge\bar\omega = 2\int_\Omega \frac{dA_z}{|H(z)|}. \tag{2.3}$$

The factor two counts both sheets. In a local conformal coordinate, the energy convention has no additional factor $1/2$.

Proof. Away from the finitely many branch points, z is a coordinate on either sheet and $\omega = f(z)\,dz$ with $|f|^2 = |H|^{-1}$. Since $(\mathrm{i}/2)\,dz\wedge d\bar z = dA_z$, each sheet contributes $\int_\Omega |H|^{-1}\,dA$. The branch points have area zero, and (2.2) gives local finiteness, so the equality extends across them. □

Proposition 2.2 (Multiplicity and local integrability)**.** *If $H(z) = (z-z_0)^k U(z)$ with $U(z_0) \neq 0$, its local reciprocal energy is finite precisely when $k = 1$ (or when $k = 0$, meaning no zero). A double zero gives logarithmic divergence; a zero of order $k > 2$ gives a power divergence.*

Proof. On a sufficiently small disk there are constants $0 < u_- \le |U| \le u_+$. The radial energy is comparable to

$$\int_0^\varepsilon r^{1-k}\,dr. \tag{2.4}$$

This converges for $k < 2$, is logarithmic for $k = 2$, and diverges more strongly for $k > 2$. The same conclusion holds on the normalization of the singular cover. For $k = 2\ell$, its two normalized

branches have $w = \pm(z - z_0)^\ell \sqrt{U}$, and ω has a pole of order ℓ. For $k = 2\ell + 1$, put $z - z_0 = \zeta^2$; then $\omega = 2\zeta^{-2\ell} U(z_0 + \zeta^2)^{-1/2}\, d\zeta$. It is regular for $k = 1$ and has a pole for each odd $k \geq 3$. These expressions also explain geometrically why the singular fiber cannot simply be included among the smooth finite-energy fibers. □

For a nonzero holomorphic field on a closed bounded domain with no boundary zero, local reciprocal integrability therefore restates simplicity of its interior zeros. We use this observation once, as a classification of the observable; it is not an independent simplicity estimate.

2.2 A disk with two real branches

In the remainder of the finite-cover analysis let

$$\Omega = D(C, R) := \{|z - C| < R\}, \quad C \in \mathbb{R}, \quad \alpha < \beta \in \Omega \cap \mathbb{R}, \tag{2.5}$$

and suppose these are exactly the two zeros of H, both simple. Then

$$H(z) = (z - \alpha)(z - \beta)V(z), \quad m = \frac{\alpha + \beta}{2}, \quad a = \frac{\beta - \alpha}{2} > 0. \tag{2.6}$$

The unit V is holomorphic and nonvanishing on the disk and its boundary. Choose a holomorphic square root of V and put

$$y_0^2 = (z - m)^2 - a^2, \quad b = V^{-1/2}, \quad \eta = \frac{dz}{y_0}, \quad \omega = b(z)\eta. \tag{2.7}$$

The two covers are identified by $w = \sqrt{V}\, y_0$.

Lemma 2.3 (Explicit annulus coordinates)**.** *The full cover is an annulus. It is parametrized by*

$$z = m + a\cosh u, \qquad y_0 = a\sinh u, \qquad u = s + \mathrm{i}\theta \pmod{2\pi\mathrm{i}}, \qquad \eta = du. \tag{2.8}$$

Writing $\delta = m - C$, its domain is $-U(\theta) \leq s \leq U(\theta)$, where

$$U(\theta) = \operatorname{arcosh}\left(\frac{-\delta\cos\theta + \sqrt{R^2 + (a^2 - \delta^2)\sin^2\theta}}{a}\right) > 0. \tag{2.9}$$

Both physical boundary components are retained.

Proof. In the nonzero coordinate $\zeta = e^u$, $z = m + (a/2)(\zeta + \zeta^{-1})$ and $y_0 = (a/2)(\zeta - \zeta^{-1})$. The inverse is $\zeta = (z - m + y_0)/a$, nonzero because $(z - m + y_0)(z - m - y_0) = a^2$. Thus these are genuine coordinates on the cover, including $u = 0, \pi\mathrm{i}$, where $y_0 = 0$ but the derivative of the pair (z, y_0) is nonzero.

The real interior roots imply $|\delta| + a < R$. Set $v = \cosh s \geq 1$. A direct calculation gives

$$|z - C|^2 = a^2 v^2 + 2a\delta v\cos\theta + \delta^2 - a^2\sin^2\theta.$$

The equation $|z - C| = R$ has the positive solution for v in (2.9). At $s = 0$, $z = m + a\cos\theta$ lies strictly inside the disk. The quadratic expression has just one crossing for $v \geq 1$, so the domain is precisely the asserted strip. Its positive continuous width and 2π periodicity identify an annulus, with boundaries $s = \pm U(\theta)$. Differentiating (2.8) gives $dz/y_0 = du$, with the identity extending at the branch points. □

We label the boundary component over $z = C + R$ with $y_0 > 0$ as the plus component. Its induced positive orientation projects counterclockwise onto the base circle; its core period for η is $2\pi\mathrm{i}$. The other component has the opposite sheet label, but its induced orientation also projects counterclockwise. The sheet involution ι satisfies $\iota^*\eta = -\eta$ and $\iota^*\omega = -\omega$.

Corollary 2.4 (Full direct coordinate formula)**.** *If $H(\bar{z}) = \overline{H(z)}$, the energy can be written without a branch singularity as*

$$\mathcal{E} = 4\int_0^{\pi}\int_0^{1} \frac{U(\theta)}{|V(m + a\cosh(tU(\theta) + \mathrm{i}\theta))|}\, dt\, d\theta. \tag{2.10}$$

Proof. In (2.8), the energy density is $|b(z)|^2\, ds\, d\theta = |V(z)|^{-1}\, ds\, d\theta$. The replacements $s \mapsto -s$ and $\theta \mapsto 2\pi - \theta$ conjugate z, while leaving U unchanged in the needed pairs. Reality of V gives the two factors of two. Finally put $s = tU(\theta)$; the area Jacobian in (s, θ) is exactly $U(\theta)$. □

3 Complete boundary identities

3.1 The complex primitive and paired cuts

Cut the annulus along a simple relative arc connecting its two physical boundary components, obtaining a disk D. Choose a single primitive F on this disk, $dF = \omega$, with one common additive constant.

Proposition 3.1 (Complete Stokes identity)**.** *With all cut edges and both physical boundaries given their induced orientations,*

$$\mathcal{E} = \frac{\mathrm{i}}{2}\int_{\partial D} F\bar{\omega}. \tag{3.1}$$

For a paired cut edge, let e_+ have its ∂D orientation and identify e_- oppositely under gluing. Then

$$\int_{e_+\cup e_-} F\bar{\omega} = J_e\overline{P_e}, \qquad J_e = F_+ - F_-, \quad P_e = \int_{e_+}\omega. \tag{3.2}$$

The jump J_e is constant; every unpaired physical arc remains in (3.1).

Proof. Holomorphicity implies $d\bar{\omega} = 0$, so $d(F\bar{\omega}) = \omega\wedge\bar{\omega}$. Stokes proves (3.1). On a paired edge the differential agrees under gluing and the orientations are opposite. Its two integrals therefore combine to $\int_{e_+}(F_+ - F_-)\bar{\omega}$. The difference has zero differential, which proves both constancy and (3.2). □

On a compact boundaryless surface, a symplectic polygon pairs all edges; the contributions are $A_j\bar{B}_j - B_j\bar{A}_j$, and hence $\mathcal{E} = \sum_j \mathrm{Im}(\bar{A}_j B_j)$. This is the usual bilinear relation [2, Section 3.1.2]. On the present annulus there are unpaired physical boundaries. It has a relative transverse path, not a compact genus-one dual cycle. Capping those boundaries without extending the differential and accounting for its poles would not justify deleting them.

3.2 A real primitive on the whole annulus

Let $q = \int_c \omega$ on the counterclockwise plus core. Let p be the integral along a fixed relative path from the plus basepoint over $z_* = C + R$ to its minus partner. Choose the path and core consistently; in the real two-root chart we take the path through β. Changing the relative path by a core adds an integer multiple of q, which leaves $\mathrm{Im}(\bar{q}p)$ unchanged.

Theorem 3.2 (Real-primitive reduction)**.** *Suppose $q \neq 0$ and define*

$$\lambda = \frac{\mathrm{i}\bar{q}}{|q|}, \quad P = \mathrm{Re}(\lambda p), \quad S = \mathrm{Im}(\bar{q}p) = -P|q|. \tag{3.3}$$

On the plus boundary, parametrized by $z = C + Re^{\mathrm{i}\theta}$, write $\omega = B(\theta)\,d\theta$ and set

$$v(\theta) = \mathrm{Re}\left(\lambda \int_0^\theta B(s)\,ds\right), \qquad t(\theta) = \mathrm{Im}(\lambda B(\theta)). \tag{3.4}$$

Then

$$\boxed{\mathcal{E} = S + R_{\mathrm{ext}}, \qquad R_{\mathrm{ext}} = 2\int_0^{2\pi} v(\theta)t(\theta)\,d\theta.} \tag{3.5}$$

If $|B| \leq K_\partial$ on the whole circle, then

$$|R_{\mathrm{ext}}| \leq 2\pi^2 K_\partial^2. \tag{3.6}$$

If $q = 0$, choose any unit λ; the same formula holds with $S = 0$ and with (3.4) defined using that choice.

Proof. Every loop in the annulus is homologous to an integer multiple of c. Since $\lambda q = \mathrm{i}|q|$, the real part of every period of $\lambda\omega$ is zero. Thus $h = \mathrm{Re}(\lambda F)$ is a single-valued harmonic function on X. Its differential changes sign under ι, so $h \circ \iota + h$ is constant. A single global change of additive constant makes h odd. At the two basepoints, $h_- - h_+ = \mathrm{Re}(\lambda p) = P$ and $h_- = -h_+$, giving $h_+ = -P/2$. Consequently the boundary traces are $h_+ = -P/2 + v$ and $h_- = -h_+$.

For completeness, in a local coordinate $x + \mathrm{i}y$ write $\lambda\omega = f\,dz$ and a primitive $h + \mathrm{i}k$. The Cauchy–Riemann equations give $h_x = \mathrm{Re}\,f$, $h_y = -\mathrm{Im}\,f$. Hence $|\nabla h|^2\,dA = (\mathrm{i}/2)\omega \wedge \bar{\omega}$. On a positively oriented boundary the outward normal flux of h is $dk = \mathrm{Im}(\lambda\omega)$. Green's formula therefore gives

$$\mathcal{E} = \int_X |\nabla h|^2\,dA = \int_{\partial X} h\ \mathrm{Im}(\lambda\omega).$$

Both sheet values change sign under ι, so the two physical components contribute equally. Using $\int_0^{2\pi} t = |q|$ yields

$$\mathcal{E} = 2\int_0^{2\pi} (-P/2 + v)t\,d\theta = -P|q| + 2\int_0^{2\pi} vt\,d\theta.$$

This proves (3.5) and its sign convention.

The function v is periodic, with $v(0) = v(2\pi) = 0$ and $|v'| \leq K_\partial$. Integrating from either endpoint gives $|v(\theta)| \leq K_\partial \min(\theta, 2\pi - \theta)$. Since the integral of this minimum is π^2, (3.6) follows. If $q = 0$, the complex primitive itself is single-valued and the preceding argument applies with $\int t = 0$, proving the stated extension. □

Remark 3.3 (What disappeared from the cut formula). The cut terms vanish for the real primitive because h is single-valued. They have not become zero information: the relative constant P remains in S. In particular, independently resetting the primitive constant on each physical boundary would destroy (3.5).

Proposition 3.4 (Real allocation between cuts and physical boundary)**.** *Choose the relative cut as above and the consistent polygon primitive. The real cut contribution to (3.1) is $S/2$, and the real physical-boundary contribution is $S/2 + R_{\mathrm{ext}}$.*

Proof. Orient one copy of the cut from minus to plus, opposite to the orientation of p. With the positive core chosen above, its primitive jump is $-q$. Equation (3.2) gives cut integral $q\bar{p}$, whose real energy contribution is $\mathrm{Re}((\mathrm{i}/2)q\bar{p}) = \mathrm{Im}(\bar{q}p)/2 = S/2$. Subtracting this contribution from (3.5) proves the second claim. An individual complex subtotal need not be real; only its real energy contribution is asserted here. □

Corollary 3.5 (An exterior floor and a finite energy interval)**.** *If $|H| \geq \mu > 0$ on the spatial circle, then $K_\partial \leq R/\sqrt{\mu}$, and*

$$\mathcal{E} \in \left[S - \frac{2\pi^2 R^2}{\mu}, S + \frac{2\pi^2 R^2}{\mu}\right] \cap [0, \infty). \tag{3.7}$$

Proof. The pullback satisfies $|B| = R/\sqrt{|H|}$; apply the theorem. □

If the field has real type, the unit is positive on the real interval, and its square root is chosen positive there, then $q = \mathrm{i}Q$ with $Q \geq 0$ (proved in Section 5). Thus $\lambda = 1$. Conjugation of the boundary trace gives $B(2\pi - \theta) = -\overline{B(\theta)}$ and $v(2\pi - \theta) = v(\theta)$, so

$$Q = 2\int_0^\pi \mathrm{Im}\, B\, d\theta, \qquad R_{\mathrm{ext}} = 4\int_0^\pi v\, \mathrm{Im}\, B\, d\theta. \tag{3.8}$$

These are exact symmetry reductions, not separate estimates of positive sectors.

4 The complete quadratic calibration

4.1 Exact energy and both boundary payments

Take $H_a(z) = z^2 - a^2$ on $|z| < R$, with $0 < a < R$. In Lemma 2.3, $m = C = 0$ and

$$U_a(\theta) = \mathrm{arsinh}\sqrt{(R/a)^2 - \cos^2\theta}, \qquad U_0 := U_a(0) = \mathrm{arcosh}(R/a). \tag{4.1}$$

Here $\omega = du$ and the full energy is the cylinder area.

Proposition 4.1 (Exact quadratic calculation)**.** *One has*

$$\mathcal{E}_a = 2\int_0^{2\pi} U_a(\theta)\, d\theta, \tag{4.2}$$

$$4\pi\, \mathrm{arcosh}(R/a) \leq \mathcal{E}_a \leq 4\pi\, \mathrm{arsinh}(R/a), \tag{4.3}$$

$$\mathcal{E}_a = 4\pi \log(2R/a) + O(a^2/R^2) \quad (a/R \longrightarrow 0). \tag{4.4}$$

With $q = 2\pi\mathrm{i}$ and $p = -2U_0$ oriented plus to minus, the real cut and physical contributions are

$$E_{\mathrm{cut}} = 2\pi U_0, \qquad E_{\mathrm{phys}} = \mathcal{E}_a - 2\pi U_0. \tag{4.5}$$

Each contains half of the leading logarithmic divergence.

Proof. The coordinate $u = s + \mathrm{i}\theta$ has energy form $(\mathrm{i}/2)\, du \wedge d\bar{u} = ds\, d\theta$, proving (4.2). The minimum and maximum of U_a are $\mathrm{arcosh}(R/a)$ and $\mathrm{arsinh}(R/a)$, proving (4.3). The identities $\mathrm{arcosh}\, x = \log(x + \sqrt{x^2 - 1})$ and $\mathrm{arsinh}\, x = \log(x + \sqrt{x^2 + 1})$ give the stated asymptotics.

To verify every boundary contribution directly, cut the cylinder at $\theta = 0$ and use $F = u$ on $0 \leq \theta \leq 2\pi$. The bottom runs from $-U_0$ to U_0, the right side upward, the top in reverse, and the left side downward. The bottom-to-top jump is $-2\pi\mathrm{i}$, so the paired cut integral is $-4\pi\mathrm{i}U_0$. Its

energy contribution is $2\pi U_0$. On the right $u = U_a + \mathrm{i}\theta$ and on the left $u = -U_a + \mathrm{i}\theta$, with opposite θ orientations. Thus

$$\begin{aligned}\int_{\partial X} F\bar{\omega} &= 2\mathrm{i}\int_0^{2\pi} (\theta U_a' - U_a)\, d\theta \\ &= 4\pi\mathrm{i}U_0 - 4\mathrm{i}\int_0^{2\pi} U_a\, d\theta.\end{aligned}$$

Multiplication by $\mathrm{i}/2$ proves (4.5). A common global primitive shift changes neither grouped contribution, since the integral of $\bar{\omega}$ over either grouped boundary pair is zero. □

For the heat control $H(\tau, z) = z^2 - 2\tau$, $a = \sqrt{2\tau}$, so

$$\mathcal{E}_\tau = 2\pi \log(1/\tau) + O(1) \qquad (\tau \downarrow 0). \tag{4.6}$$

The finite core period is not a finite-energy endpoint statement.

4.2 A scoped limitation of bounded exterior summaries

Proposition 4.2. *Fix $R > 0$ and $0 < a \le R/2$. The core period, an exterior field floor, every prescribed finite collection of exterior differential derivative upper bounds, and separately normalized boundary primitive oscillation upper bounds can all be chosen uniformly in a. Nevertheless $\mathcal{E}_a$ is unbounded as $a \downarrow 0$. No finite upper estimate for the energy can depend only on those fixed common budget inequalities.*

Proof. The core period is identically $2\pi\mathrm{i}$. On $|z| = R$, $|H_a| \ge 3R^2/4$. On a fixed collar, for example $3R/4 \le |z| \le 5R/4$, the expression

$$\frac{1}{\sqrt{z^2 - a^2}} = \frac{1}{z}\left(1 - \frac{a^2}{z^2}\right)^{-1/2}$$

is holomorphic with the branch near one and uniformly bounded. Cauchy's estimates on smaller fixed collars bound every fixed finite derivative collection. In the cylinder parametrization the separately rebased boundary primitives have increments $\pm(U_a(\theta) - U_0) + \mathrm{i}\theta$; they are uniformly bounded since

$$0 \le U_a - U_0 \le \operatorname{arsinh}(R/a) - \operatorname{arcosh}(R/a).$$

But (4.3) diverges. The missing relative constant is $2U_0$, which is unbounded. □

The scope is important. Exact attained norms or complete exact boundary values can encode a and permit its recovery. The proposition concerns fixed common upper-budget inequalities, not every possible use of exterior data. It also shows why an absolute field error set admitting arbitrarily small positive a admits arbitrarily large finite reciprocal energy, even if each admitted positive-a fiber is regular.

4.3 A lower bound for general pair charts

Lemma 4.3. *Suppose $H(z) = (z^2 - a^2)V(z)$ on $|z| \le R$, $0 < 2a < R$, and $|V| \le M$. Then*

$$\mathcal{E} \ge \frac{16\pi}{5M} \log \frac{R}{2a}. \tag{4.7}$$

Consequently, an independently obtained upper $\mathcal{E} \le B$ implies $a \ge (R/2)\exp[-5MB/(16\pi)]$.

Proof. On $2a \le |z| \le R$, $|z^2 - a^2| \le (5/4)|z|^2$. Integrating $2 \cdot 4/(5M|z|^2)$ over this annulus proves (4.7); rearranging proves the consequence. □

An endpoint conclusion requires a fixed positive R and uniform M, B. The upper estimates later in this manuscript do not supply uniform B.

4.4 Capacity as supporting analysis

On the round annulus $r < |z| < R$, the function $\varphi(z) = \log(|z|/r)/\log(R/r)$ has boundary values zero and one. Direct polar integration gives, with no factor $1/2$,

$$\operatorname{cap} A(r, R) = \int |\nabla\varphi|^2 \, dA = \frac{2\pi}{\log(R/r)}.$$

For a closed real one-form of core period p, circlewise Cauchy–Schwarz and radial integration give

$$\int |\alpha|^2 \, dA \ge \frac{p^2}{2\pi} \log(R/r).$$

Indeed its tangential integral is p on each circle, so its energy on that circle is at least $p^2/(2\pi s)$. These formulas explain the degeneration scale but do not give an upper modulus bound. In particular, a trial potential gives an upper capacity bound and hence a lower bound for the fixed-period core energy, not the upper estimate sought here.

5 Quadratic subtraction with a bounded full remainder

Continue with the disk and unit factor in (2.5)–(2.7). Assume throughout this section that

$$|V| \ge \nu > 0, \qquad |b'| \le L, \qquad |b''| \le K \quad \text{on } \overline{D(C, R)}, \qquad b = V^{-1/2}. \tag{5.1}$$

These are explicit budgets on a fixed disk. Uniformity below means independence of the branch separation within a family for which the same R, ν, L, K work. Section 6 derives such budgets from the complete canonical source on an event-dependent chart.

5.1 The mean core coefficient and an exact primitive

Lemma 5.1 (The core mean)**.** *With the plus marking of Section 2,*

$$c := \frac{q}{2\pi\mathrm{i}} = \frac{1}{\pi} \int_0^\pi b(m + a\cos\theta) \, d\theta. \tag{5.2}$$

Writing $b_0 = b(m)$, one has

$$|c|, |b_0| \le \nu^{-1/2}, \qquad |c - b_0| \le \frac{Ka^2}{4}. \tag{5.3}$$

If V is real and nonzero on the slit $[\alpha, \beta]$, then $c \ne 0$.

Proof. Shrink the plus contour counterclockwise onto the slit. On its upper bank $y_0 = \mathrm{i}\sqrt{a^2 - (z-m)^2}$ and the direction is from β to α; the lower bank has the opposite square root and direction. Their contributions add to

$$q = 2\mathrm{i} \int_\alpha^\beta \frac{b(x)}{\sqrt{a^2 - (x-m)^2}} \, dx.$$

The endpoint arcs vanish in the local regular coordinate (2.2). Substituting $x = m + a\cos\theta$ proves (5.2). The bound for c follows by averaging $|b|$. Taylor's formula on the real segment gives

$$b(m+h) = b_0 + hb'(m) + h^2 \int_0^1 (1-t)b''(m+th)\,dt.$$

The mean of $\cos\theta$ is zero and that of $\cos^2\theta$ is $1/2$, proving the second estimate. For real nonzero V, its sign is constant on the slit. The coherent b has one fixed phase times a strictly positive real function there, so its average cannot vanish. For a general complex unit no such nonvanishing assertion is made. □

Proposition 5.2 (Exact quadratic subtraction)**.** *There is a single-valued holomorphic function G on the whole cover, normalized by $G(\beta) = 0$, such that*

$$\omega = c\eta + dG, \qquad G = y_0 f(z), \tag{5.4}$$

where f is holomorphic on the base disk and satisfies

$$((z-m)^2 - a^2)f'(z) + (z-m)f(z) = b(z) - c. \tag{5.5}$$

If $\mathcal{E}_0$ is the full energy of η, then

$$\mathcal{E} = |c|^2\mathcal{E}_0 + 2\,\mathrm{Re}\left[\bar{c}\frac{\mathrm{i}}{2}\int_{\partial X} G\bar{\eta}\right] + \frac{\mathrm{i}}{2}\int_{\partial X} G\,d\bar{G}. \tag{5.6}$$

Here ∂X consists of the two physical boundaries.

Proof. The only period of $\omega - c\eta$ is $q - 2\pi\mathrm{i}c = 0$. Its path integral is therefore a single-valued holomorphic primitive G. The differential changes sign under sheet interchange, so $G \circ \iota + G$ is constant. Its value at the fixed point β is zero, proving that G is odd and, in particular, also vanishes at α. The quotient G/y_0 is even and descends away from the two branches. At either branch use $z - \gamma = \zeta^2$. Both G and y_0 are odd in ζ, while y_0/ζ is nonzero. Their quotient extends holomorphically. Differentiation of $G = y_0 f$ proves (5.5).

Expand the energy of $c\eta + dG$. The two mixed integrals are complex conjugates. For one of them, $d(G\bar{\eta}) = dG \wedge \bar{\eta}$, and the last term is $d(G\,d\bar{G}) = dG \wedge d\bar{G}$. Stokes gives (5.6). These correction forms are single-valued; paired cut edges cancel, so no relative or physical contribution was suppressed. The mixed term in (5.6) is essential. □

5.2 Uniform energy estimates

Set $\delta = m - C$, $R_- = R - |\delta|$, $R_+ = R + |\delta|$. Disk inclusion about m and (4.3) give

$$4\pi\,\mathrm{arcosh}(R_-/a) \le \mathcal{E}_0 := 2\int_{D(C,R)} \frac{dA}{|z-\alpha||z-\beta|} \le 4\pi\,\mathrm{arsinh}(R_+/a). \tag{5.7}$$

The roots are interior, so $R_- > a$ and $R_+ < 2R$.

Theorem 5.3 (Bounds independent of the gap)**.** *Under (5.1), define*

$$C_{\mathrm{osc}} = \frac{8\pi RL}{\sqrt{\nu}} + 8\pi R^2 L^2, \tag{5.8}$$

$$C_E = C_{\mathrm{osc}} + \frac{8\pi K R^2}{\sqrt{\nu}}. \tag{5.9}$$

Then the complete energy satisfies

$$|\mathcal{E} - |b_0|^2\mathcal{E}_0| \le C_{\mathrm{osc}}, \qquad |\mathcal{E} - |c|^2\mathcal{E}_0| \le C_E. \tag{5.10}$$

Proof. For any γ in the disk, let $\rho(\theta)$ be the exit distance along the ray from γ in direction θ. Polar integration and Cauchy–Schwarz give

$$\int_{D(C,R)} \frac{dA}{|z-\gamma|} = \int_0^{2\pi} \rho(\theta)\, d\theta \le \sqrt{2\pi \int_0^{2\pi} \rho(\theta)^2\, d\theta} = 2\pi R.$$

Put $d_1 = |z-\alpha|$, $d_2 = |z-\beta|$. Since $|z-m| \le (d_1+d_2)/2$ and $|z-m| < 2R$,

$$I_1 := \int \frac{|z-m|}{d_1 d_2}\, dA \le 2\pi R, \qquad I_2 := \int \frac{|z-m|^2}{d_1 d_2}\, dA \le 4\pi R^2. \tag{5.11}$$

All integrals here are over the full base disk. The straight segment from m to z remains in the disk, so $|b(z)-b_0| \le L|z-m|$. Expanding the square, including its mixed term, yields

$$\big||b(z)|^2 - |b_0|^2\big| \le 2\nu^{-1/2} L|z-m| + L^2|z-m|^2.$$

Multiply by $2/(d_1 d_2)$ and apply (5.11); this proves the first bound. By (5.3),

$$\big||c|^2 - |b_0|^2\big| \le \frac{Ka^2}{2\sqrt{\nu}}.$$

Furthermore $\operatorname{arsinh} x \le x$ for $x \ge 0$, whence $a^2 \operatorname{arsinh}(R_+/a) \le aR_+ \le R_+^2 \le 4R^2$. Using the upper bound in (5.7) proves the second estimate with exactly the constant (5.9). □

5.3 The relative period and the full boundary remainder

Let $\ell = C + R - m > a$. For the path from plus to minus through β,

$$p = -2\int_a^{\ell} \frac{b(m+y)}{\sqrt{y^2-a^2}}\, dy, \qquad p_0 = -2\operatorname{arcosh}(\ell/a). \tag{5.12}$$

The negative sign records travel toward the branch on the plus sheet; the return on the minus sheet contributes the same integral.

Theorem 5.4 (Complete transverse and boundary bounds)**.** *Put $C_p = 4LR + 2KR^2$. With the real contributions of Proposition 3.4,*

$$|p - cp_0| \le C_p, \tag{5.13}$$

$$|E_{\mathrm{cut}} + \pi|c|^2 p_0| \le \pi|c| C_p, \tag{5.14}$$

$$|E_{\mathrm{phys}} - |c|^2(\mathcal{E}_0 + \pi p_0)| \le C_E + \pi|c| C_p. \tag{5.15}$$

These bounds retain the full physical boundary, not only a selected bounded subtotal.

Proof. By (5.12) and $|b(m+y) - b_0| \le Ly$,

$$|p - b_0 p_0| \le 2L\int_a^{\ell} \frac{y}{\sqrt{y^2-a^2}}\, dy = 2L\sqrt{\ell^2-a^2} \le 4LR.$$

Since $\operatorname{arcosh} x \le x$ for $x \ge 1$,

$$|b_0 - c||p_0| \le \frac{K}{2} a^2 \operatorname{arcosh}(\ell/a) \le \frac{K}{2}\ell^2 \le 2KR^2.$$

This proves (5.13). Now $q = 2\pi i c$ gives $E_{\mathrm{cut}} = \operatorname{Im}(\bar{q}p)/2 = -\pi \operatorname{Re}(\bar{c}p)$. Subtract the corresponding model to prove (5.14). Finally $E_{\mathrm{phys}} = \mathcal{E} - E_{\mathrm{cut}}$; Theorem 5.3 and (5.14) prove (5.15). □

Both model boundary payments still have logarithmic growth. The bounded remainders locate that growth precisely and keep the algebra complete; they do not make the endpoint energy finite. In particular, boundedness of G in an unquantified family is neither assumed nor needed in the proof of the estimates.

6 A complete moment-controlled heat-event chart

The preceding estimates apply to a holomorphic pair chart. We now derive all of its constants from the complete canonical source at an assumed ordinary double. This construction is conditional local analysis: it does not assert the existence of such an event for the theta field.

6.1 Complete-source analyticity and derivative bounds

Put $\lambda_n = \pi n^2$. The nth summand in (1.2) is

$$\Phi_n(u) = \lambda_n e^{5u}(2\lambda_n e^{4u} - 3)e^{-\lambda_n e^{4u}} > 0 \qquad (u \geq 0),$$

because $2\lambda_n e^{4u} - 3 \geq 2\pi - 3 > 0$. For the positive leading terms the successive ratio satisfies

$$\frac{(n+1)^4 e^{-\pi(n+1)^2 e^{4u}}}{n^4 e^{-\pi n^2 e^{4u}}} \leq 16e^{-3\pi} =: q_0 < 1.$$

Dropping the negative part only for this upper bound and summing the geometric majorant gives

$$0 < \Phi(u) \leq \frac{2\pi^2}{1-q_0} e^{9u - \pi e^{4u}}. \tag{6.1}$$

Consequently, for every real $S, Y \geq 0$, and nonnegative integer j,

$$M_{j,Y}(S) := \int_0^\infty u^j e^{Su^2 + Yu} \Phi(u)\, du < \infty. \tag{6.2}$$

The decay of $e^{-\pi e^{4u}}$ dominates every displayed quadratic, linear, and polynomial factor. This also gives a common integrable majorant on every compact subset of the two complex variables (t, z). Thus (1.1) defines a jointly entire function, its derivatives can be taken under the complete integral, and

$$|\partial_z^j H(t,z)| \leq M_{j,Y}(S) \quad \text{if } \operatorname{Re} t \leq S, \quad |\operatorname{Im} z| \leq Y. \tag{6.3}$$

Indeed the derivatives of cosine are cosine or sine with the factor u^j, and both trigonometric functions have modulus at most $e^{|\operatorname{Im} z| u}$. Time differentiation adds two powers of u and gives (1.4). The source is real, so $H(t,z)$ is real for real t, z. These statements use all modes and the whole half-line. An explicit finite acquisition ledger for the moments is given in Appendix B.

6.2 The finite ordinary-double chart

Assume

$$T \geq 0, \quad x_* \in \mathbb{R}, \quad H(T, x_*) = H_z(T, x_*) = 0, \qquad A = \tfrac{1}{2} H_{zz}(T, x_*) \in \mathbb{R} \setminus \{0\}. \tag{6.4}$$

Use the shifted field and complete moments

$$F_\tau(x) = H(T + \tau, x_* + x), \quad h(x) = F_0(x), \quad M_j = M_{j,1}(T+1) \quad (j \geq 0). \tag{6.5}$$

On the complex bidisk $|\tau| \leq 1$, $|x| \leq 1$, formula (6.3) bounds the jth spatial derivative of F_τ by M_j. In particular the following constants are finite and strictly positive:

$$r = \min\{1, |A|/(4M_3)\}, \tag{6.6}$$

$$\sigma = \min\{1, 3|A|r^2/(8M_2), |A|r/(4M_3), |A|/(4M_4)\}. \tag{6.7}$$

They depend on the event and its curvature. No lower bound uniform in height, time, or the event is asserted.

Theorem 6.1 (Explicit finite event chart)**.** *For every complex $|\tau| \le \sigma$, the field F_τ has exactly two zeros in $|x| < r$, counted with multiplicity, and obeys the outer floor*

$$|F_\tau(x)| \ge m_{\text{out}} := 3|A|r^2/8 \qquad (|x| = r). \tag{6.8}$$

There is exactly one critical point $k(\tau)$ in this disk. It depends holomorphically on τ, and the two zeros are simple for $0 < |\tau| \le \sigma$. For real $0 < \tau \le \sigma$ they are real and can be ordered as $\alpha < k(\tau) < \beta$. With

$$m = \frac{\alpha+\beta}{2}, \qquad a = \frac{\beta-\alpha}{2} > 0, \tag{6.9}$$

one has

$$\frac{6\tau}{5} \le a^2 \le \frac{10\tau}{3}, \qquad |m| \le \frac{28M_3}{27|A|}\tau. \tag{6.10}$$

In particular the finite physical cover is a smooth bordered annulus for each such positive time, with both physical boundaries retained.

Proof. Taylor's formula with integral remainder gives $h(x) = x^2 g(x)$ and

$$|g(x) - A| \le M_3|x|/6 \le |A|/24 \qquad (|x| \le r). \tag{6.11}$$

Thus g is nonvanishing on the closed disk. The only zero of h there is its double zero at the origin, and on the circle $|h| \ge 3|A|r^2/4$. Integrating $\partial_\tau F_\tau = -F_\tau''$ along the complex time segment gives

$$|F_\tau(x) - h(x)| \le M_2|\tau| \le 3|A|r^2/8. \tag{6.12}$$

The outer floor follows by subtraction. Since the right side is strictly less than the boundary lower bound for $|h|$, Rouché's theorem gives the count two.

On the entire closed disk and time window, spatial and time integration of the derivative bounds gives

$$|F_\tau''(x) - 2A| \le M_3 r + M_4\sigma \le |A|/2. \tag{6.13}$$

On $|x| = r$, Taylor's theorem in space and $\partial_\tau F_\tau' = -F_\tau'''$ likewise give

$$|F_\tau'(x) - 2Ax| \le M_3 r^2/2 + M_3\sigma \le 3|A|r/8 < 2|A|r.$$

Rouché's theorem applied to F_τ' gives exactly one critical point, counted with multiplicity. Its second derivative is nonzero by (6.13). The local implicit functions therefore patch, by uniqueness, to one holomorphic function $k(\tau)$ on the time disk, with $k(0) = 0$.

Set $v(\tau) = F_\tau(k(\tau))$. The chain rule, the critical-point equation, and the heat equation imply

$$v' = -F_\tau''(k(\tau)), \quad v(0) = 0, \quad |v(\tau) + 2A\tau| \le |A||\tau|/2. \tag{6.14}$$

The last inequality follows by integrating (6.13) along the complex segment. It implies $|v(\tau)| \ge 3|A||\tau|/2 > 0$ when $\tau \ne 0$. A multiple zero would have to be the unique critical point, and hence would force $v = 0$. Thus both zeros are simple throughout the punctured time disk.

Now let τ be positive and real. Reality and uniqueness give a real critical point. On the real spatial interval the function $f = F_\tau/A$ satisfies

$$3/2 \le f'' \le 5/2, \qquad -5\tau/2 \le f(k) \le -3\tau/2. \tag{6.15}$$

The required endpoint signs follow from more than the modulus floor: (6.11) gives $h(\pm r)/A \ge 3r^2/4$, and (6.12) gives $|F_\tau(\pm r) - h(\pm r)|/|A| \le 3r^2/8$. Hence $f(\pm r) \ge 3r^2/8 > 0$. Strict convexity now yields one real zero on each side of k, and these exhaust the count two.

For either root γ, Taylor's theorem about k gives

$$\tfrac{3}{4}|\gamma - k|^2 \le -f(k) \le \tfrac{5}{4}|\gamma - k|^2.$$

Combining this with (6.15), each of the distances $k - \alpha$ and $\beta - k$ lies between $\sqrt{6\tau/5}$ and $\sqrt{10\tau/3}$. Their arithmetic mean is a, proving its two-sided estimate in (6.10).

To control the midpoint, first use $|F_\tau'(0)| \le M_3\tau$, which follows from the time derivative of F_τ' and the event equation. The real curvature floor then gives

$$|k| \le \frac{2M_3}{3|A|}\tau.$$

Expand $F_\tau(m + a) = F_\tau(m - a) = 0$ through second order at m. The two cubic remainders have magnitudes at most $M_3a^3/6$ each. Subtracting the expansions gives $|F_\tau'(m)| \le M_3a^2/6$. Another application of the curvature floor yields

$$|m - k| \le \frac{M_3a^2}{9|A|} \le \frac{10M_3}{27|A|}\tau.$$

Adding the last two estimates proves the midpoint bound. Finally the outer floor and simple zero count give the regular finite cover, and Lemma 2.3 identifies it as the asserted annulus. □

6.3 Deflation and derivative constants from the same source

For the remainder of this section $0 < \tau \le \sigma$ is real and $F_\tau(x) = (x - \alpha)(x - \beta)V(x)$.

Lemma 6.2 (Explicit deflated-unit estimates)**.** *On the complex disk $|x| \le r$, set $\nu = 3|A|/4$. Then*

$$|V - A| \le |A|/4, \qquad |V| \ge \nu, \tag{6.16}$$

$$|V'| \le M_3/6, \qquad |V''| \le M_4/12. \tag{6.17}$$

For a coherent holomorphic choice $b = V^{-1/2}$ the constants

$$L = \frac{M_3}{12\nu^{3/2}}, \qquad K = \frac{M_3^2}{48\nu^{5/2}} + \frac{M_4}{24\nu^{3/2}} \tag{6.18}$$

satisfy $|b| \le \nu^{-1/2}$, $|b'| \le L$, and $|b''| \le K$. At the root midpoint one has the sharper estimate

$$|V(m) - A| \le J\tau, \qquad J = \frac{14M_3^2}{27|A|} + \frac{23M_4}{36}. \tag{6.19}$$

Proof. Let $\Delta = \{(s, t) : s, t \ge 0,\ s + t \le 1\}$. Twice applying the fundamental theorem of calculus along complex line segments gives

$$V(x) = \int_\Delta F_\tau''(s\alpha + t\beta + (1 - s - t)x)\, ds\, dt. \tag{6.20}$$

To verify the normalization, for distinct x, α, β first integrate in t and then in s. The result is

$$\frac{1}{\beta - x}\left\{\frac{F_\tau(\alpha) - F_\tau(\beta)}{\alpha - \beta} - \frac{F_\tau(\alpha) - F_\tau(x)}{\alpha - x}\right\} = \frac{F_\tau(x)}{(x - \alpha)(x - \beta)}.$$

The cases $x = \alpha, \beta$ follow by removable continuation. The disk is convex, so every evaluation point in (6.20) belongs to the same disk on which our source derivative bounds hold. The elementary simplex integrals are

$$\int_\Delta 1 = \tfrac{1}{2}, \qquad \int_\Delta (1-s-t) = \tfrac{1}{6}, \qquad \int_\Delta (1-s-t)^2 = \tfrac{1}{12},$$
$$\int_\Delta (t-s) = 0, \qquad \int_\Delta (t-s)^2 = \tfrac{1}{12}.$$

For example these follow by successive integration of monomials, or from $\int_\Delta s^i t^j (1-s-t)^k \, ds\, dt = i!j!k!/(i+j+k+2)!$. Integrating (6.13) proves (6.16); differentiating (6.20) once and twice in x and using the next two simplex integrals proves (6.17).

The unit has no zero, so it has a holomorphic square root on the disk. The derivative identities

$$b' = -\tfrac{1}{2} V^{-3/2} V', \qquad b'' = \tfrac{3}{4} V^{-5/2} (V')^2 - \tfrac{1}{2} V^{-3/2} V''$$

give exactly (6.18). On the real interval, V is real and (6.16) makes its sign the sign of A. Therefore b has a fixed nonzero phase there. In particular its arcsine average, which determines the core period in Section 5, is nonzero.

At $x = m$ the simplex evaluation point is $m + a(t-s)$. Expand F_τ'' at m. Its linear term integrates to zero, and the quadratic remainder and the last simplex integral give

$$|V(m) - \tfrac{1}{2} F_\tau''(m)| \le M_4 a^2/24.$$

Time integration and spatial integration also give $|F_\tau''(m) - 2A| \le M_4 \tau + M_3 |m|$. Combining with (6.10) yields

$$\begin{aligned} |V(m) - A| &\le \tfrac{1}{2} M_4 \tau + \tfrac{1}{2} M_3 |m| + M_4 a^2/24 \\ &\le \left(\frac{14 M_3^2}{27|A|} + \frac{1}{2} M_4 + \frac{5}{36} M_4 \right) \tau, \end{aligned}$$

which is (6.19). □

The deflated-unit floor (6.16) is a conclusion of the event equations and complete derivative bounds. It is not a lower floor for F_τ, which vanishes at each branch. Theorem 5.3 therefore applies on this chart with the explicitly derived constants ν, L, K, r; its constants (5.8) and (5.9) do not depend on the shrinking pair separation.

6.4 A finite coarse upper bound

Proposition 6.3. *For $0 < \tau \le \sigma$ the complete energy obeys*

$$\mathcal{E}_\tau \le B(\tau) := \frac{8}{3|A|} \left[4\pi + 2\pi \log \left(2r \sqrt{\frac{5}{6\tau}} \right) \right]. \tag{6.21}$$

In particular $[0, B(\sigma/2)]$ encloses all energies on the bounded window $\sigma/2 \le \tau \le \sigma$.

Proof. Set $d = \beta - \alpha = 2a$. Split the base disk into its intersections with $B(\alpha, d/2)$ and $B(\beta, d/2)$ and the complement. The two small disks are disjoint except at their tangent point. On the first, $|x - \beta| \ge d/2$, so its contribution to the integral of $1/(|x-\alpha||x-\beta|)$ is bounded by the integral of $2/(d|x-\alpha|)$ over the full radius-$d/2$ disk, which is 2π. The same bound holds at the other root.

On the complement the two distances p, q lie between $d/2$ and $2r$. Use $1/(pq) \le (p^{-2}+q^{-2})/2$ and bound each resulting integral by its full centered annulus integral. This gives

$$\int_{|x|<r} \frac{dA_x}{|x-\alpha||x-\beta|} \le 4\pi + 2\pi \log(4r/d).$$

The two-sheet energy has the additional factor two, and $|V| \ge 3|A|/4$. Finally $d \ge 2\sqrt{6\tau/5}$ by (6.10). These facts give (6.21). The right side decreases with τ, proving the window statement. □

6.5 A strictly sharper shared upper certificate

Write the explicit constant (5.8) on the present chart as

$$C_{\mathrm{osc}} = 8\pi r L/\sqrt{\nu} + 8\pi r^2 L^2. \tag{6.22}$$

Define

$$D = 2r\sqrt{5/6}, \qquad c_* = \log 2 + \tfrac{1}{4}, \tag{6.23}$$

$$K_* = \max\{0, 3|A|C_{\mathrm{osc}}/(2\pi) + 7c_* - 16\}, \tag{6.24}$$

$$s_* = \min\{\sigma, D^2, \nu/(6J), D^2 e^{-2(K_*+1)}\} > 0. \tag{6.25}$$

The canonical source has $J > 0$, since its moments are positive. If the same argument is used for a polynomial calibration with $J = 0$, the division term in (6.25) is omitted.

Theorem 6.4 (Strict improvement of the coarse certificate)**.** *For $0 < \tau \le s_*$ a valid complete upper bound is*

$$\mathcal{E}_\tau \le U(\tau) := 4\pi\left(\frac{1}{|A|} + \frac{J\tau}{|A|\nu}\right)\operatorname{arsinh}(D/\sqrt{\tau}) + C_{\mathrm{osc}}. \tag{6.26}$$

It satisfies the explicit strict comparison

$$B(\tau) - U(\tau) \ge \frac{2\pi}{3|A|} > 0. \tag{6.27}$$

Thus $[0, U(\tau)]$ is strictly contained in $[0, B(\tau)]$ on this declared window.

Proof. The root midpoint is real and the two roots are interior, so $r - |m| > a$ and $r + |m| \le 2r$. The upper shifted-disk bound (5.7) gives

$$\mathcal{E}_0 \le 4\pi \operatorname{arsinh}((r+|m|)/a) \le 4\pi \operatorname{arsinh}(D/\sqrt{\tau}),$$

where the last step uses (6.10). Also (6.16) and (6.19) imply

$$\frac{1}{|V(m)|} - \frac{1}{|A|} = \frac{|A| - |V(m)|}{|A||V(m)|} \le \frac{J\tau}{|A|\nu}.$$

Apply the $b(m)$ version of Theorem 5.3, with remainder (5.8), to obtain (6.26).

For $x \ge 1$ the elementary inequalities $\sqrt{1+y} \le 1 + y/2$ and $\log(1+y/4) \le y/4$, with $y = x^{-2} \le 1$, give

$$\operatorname{arsinh} x = \log x + \log(1 + \sqrt{1+x^{-2}}) \le \log x + \log 2 + \tfrac{1}{4}. \tag{6.28}$$

Let $\ell = \log(D/\sqrt{\tau})$. From (6.25), $\ell \ge K_* + 1$, $D/\sqrt{\tau} \ge 1$, and $1 + J\tau/\nu \le 7/6$. The coarse certificate is $B = (2\pi/(3|A|))(16 + 8\ell)$, whereas (6.26) and (6.28) give

$$U \le \frac{2\pi}{3|A|}(7\ell + 7c_*) + C_{\mathrm{osc}}.$$

Subtracting gives

$$B - U \geq \frac{2\pi}{3|A|}\left[\ell + 16 - 7c_* - \frac{3|A|C_{\mathrm{osc}}}{2\pi}\right] \geq \frac{2\pi}{3|A|},$$

by the definition of K_*. □

This is a strict improvement over a specified coarse certificate. Both bounds are available to direct integration supplied with the same roots, unit, and source bounds. Thus the theorem gives a sharper shared direct baseline; it supplies no ordering of all boundary and direct methods.

6.6 The resulting degeneration is still logarithmic

Corollary 6.5 (Ordinary-double energy scale)**.** *For the fixed event chart above,*

$$\mathcal{E}_\tau = \frac{2\pi}{|A|}\log(1/\tau) + O(1) \qquad (\tau \downarrow 0). \tag{6.29}$$

The bounded remainder may depend on the event and its chart constants.

Proof. By (6.10), $a \longrightarrow 0$, $\log(1/a) = \frac{1}{2}\log(1/\tau) + O(1)$, and $m = O(\tau)$. In particular, for sufficiently small time both $r - |m|$ and $r + |m|$ lie in a fixed positive interval about r. Apply the lower and upper shifted-disk bounds (5.7) and the elementary large-argument asymptotics used in (4.4). They give

$$\mathcal{E}_0 = 4\pi\log(1/a) + O(1) = 2\pi\log(1/\tau) + O(1).$$

The unit estimate (6.19) and the floor $|V(m)| \geq \nu$ give the two-sided estimate

$$\left|\frac{1}{|V(m)|} - \frac{1}{|A|}\right| \leq \frac{J\tau}{|A|\nu}.$$

Finally Theorem 5.3, using (5.8), gives $\mathcal{E}_\tau = |V(m)|^{-1}\mathcal{E}_0 + O(1)$. The normalization error is $O(\tau\log(1/\tau))$, hence bounded and tending to zero. This proves (6.29). □

The positive logarithmic coefficient is consistent with the exact quadratic control. Neither (6.21) nor (6.26) remains bounded as τ tends to the singular fiber. A source estimate that excludes the event would have to add information not supplied by this generic ordinary-double chart.

7 Analytic consequences for finite enclosures

The preceding statements are exact. A numerical acquisition must also control the uncertainty of every term that it uses. This section records the analytic bounds behind such an acquisition; it does not prescribe an implementation or assume that the boundary expression is cheaper.

7.1 Primitive products and period products

Suppose the cut polygon is decomposed into finitely many oriented arcs with real parameters s. On each arc write $\omega = B(s)\,ds$ and let F denote the coherent primitive in Proposition 3.1. Suppose approximations satisfy

$$|F - \widehat{F}| \leq \delta_F(s), \qquad |B - \widehat{B}| \leq \delta_B(s).$$

An independent quadrature enclosure for $(\mathrm{i}/2)\sum\int \widehat{F}\overline{\widehat{B}}\,ds$ has error E_{quad}. Then its real part $\widehat{E}$ satisfies

$$|\mathcal{E} - \widehat{E}| \leq E_{\mathrm{quad}} + \frac{1}{2}\sum\int\left(\delta_F|\widehat{B}| + |\widehat{F}|\delta_B + \delta_F\delta_B\right)\,ds. \tag{7.1}$$

Indeed expand $(\widehat{F}+\Delta_F)\overline{(\widehat{B}+\Delta_B)}$ and apply the triangle inequality. The primitive error includes the basepoint and accumulated integral errors; it cannot be reset independently on different boundary arcs. Paired-edge cancellation is usable only with the corresponding common primitive and jump enclosure.

Likewise, if $|q-\widehat{q}| \le \epsilon_q$ and $|p-\widehat{p}| \le \epsilon_p$, then

$$\left|\mathrm{Im}(\bar{q}p) - \mathrm{Im}(\overline{\widehat{q}}\widehat{p})\right| \le |\widehat{q}|\epsilon_p + |\widehat{p}|\epsilon_q + \epsilon_q\epsilon_p. \tag{7.2}$$

This follows by the same expansion. In a collision chart $|p|$ grows logarithmically, so a fixed absolute uncertainty in q is amplified. A bounded exact remainder in Theorem 5.4 alone does not imply a bounded acquisition error.

7.2 Simple-root neighborhoods

If $H(z) = (z-\alpha)U(z)$ on $|z-\alpha| \le \eta$, with $0 < \nu_1 \le |U| \le M_1$, then direct polar integration gives

$$\frac{4\pi\eta}{M_1} \le \mathcal{E}(H; D(\alpha,\eta)) \le \frac{4\pi\eta}{\nu_1}. \tag{7.3}$$

Thus the neighborhood of a simple zero has an explicit finite payment. Equivalently the coordinate (2.2) removes the local singularity. An exterior floor on H is never used as an interior floor across a root. When two roots are present, the full coordinate formula (2.10) is a direct integration route using exactly the same unit and root information as the period formulas.

7.3 A complete analytic approximation and exact deflation

Let H be holomorphic on $|z-C| \le \mathcal{R}$ with $|H| \le M$ there, and let $0 < R < \mathcal{R}$, $\rho = R/\mathcal{R}$. For the degree-N Taylor polynomial P_N about C, Cauchy's coefficient estimate gives, when $N \ge 2$,

$$\epsilon_0 := \sup_{|z-C|\le R} |H - P_N| \le M\frac{\rho^{N+1}}{1-\rho}, \tag{7.4}$$

$$\epsilon_1 := \sup_{|z-C|\le R} |H' - P_N'| \le \frac{M}{\mathcal{R}}\rho^N\left(\frac{N+1}{1-\rho} + \frac{\rho}{(1-\rho)^2}\right), \tag{7.5}$$

$$\epsilon_2 := \sup_{|z-C|\le R} |H'' - P_N''| \le \frac{M}{\mathcal{R}^2}\rho^{N-1}\left(\frac{N(N+1)}{1-\rho} + \frac{(2N+1)\rho}{(1-\rho)^2} + \frac{\rho(1+\rho)}{(1-\rho)^3}\right). \tag{7.6}$$

For example the last formula is the sum of $(N+1+k)(N+k)\rho^k$, using $\sum \rho^k = (1-\rho)^{-1}$, $\sum k\rho^k = \rho(1-\rho)^{-2}$, and $\sum k^2\rho^k = \rho(1+\rho)(1-\rho)^{-3}$. The first two are the corresponding zeroth and first derivatives of the geometric tail. For the complete theta field, M can be supplied by a complete source moment; (1.3) is also an exact alternative at $t = 0$.

Let α, β be the exact simple roots, enclosed independently, and define the polynomial divided difference $Q_N(z) = P_N[\alpha, \beta, z]$. Even though P_N need not vanish at the exact roots, the complete function does vanish at them, and its divided difference is $V(z) = H[\alpha, \beta, z]$. The elementary simplex identity proved in Section 6 therefore yields

$$|V(z) - Q_N(z)| \le \frac{\epsilon_2}{2} \quad (z, \alpha, \beta \in \overline{D(C,R)}). \tag{7.7}$$

Indeed the second derivative of $H - P_N$ is integrated over a simplex of area $1/2$; the disk is convex. This argument does not divide a field error by $|(z-\alpha)(z-\beta)|$ near a root. It retains the exact

removable continuation and a uniform error bound. Root enclosures must still be propagated through Q_N and the coordinate map; replacing the roots by their midpoints would not justify (7.7).

All these bounds apply before selecting a mesh or an arithmetic package. They specify the error information required by either comparison route. In particular, taking the intersection of two proved intervals is a shared improvement available to both routes.

8 Recorded numerical support

The finite analytic framework has been applied to one actual regular theta fiber, without an RH assumption. The recorded outward-ball calculation [5], using Arb arithmetic [3], treats $t = 0$ and $|z - 35| \le 8$. A complete boundary winding count certifies exactly two interior roots, each separately enclosed as simple and real:

$$\alpha \simeq 28.269450283469387580914503967 12, \qquad \beta \simeq 42.044079277543109985256959 18780.$$

The certificate uses the exact normalization (1.3), with complete analytic Taylor and divided-difference tails, rather than a finite source head treated as the full field. Its native boundary interval is

$$[28770927.163097,\ 33752291.350598], \tag{8.1}$$

and its native direct-coordinate interval is

$$[28728270.232214,\ 34655897.029090]. \tag{8.2}$$

The endpoints are outward rational bounds as printed. Both intervals meet the recorded common relative-diameter target, and the first lies strictly inside the second. The result supports a finite acquisition on a genuine regular fiber. It compares the two specified native implementations; a direct method permitted to reuse all the same identities and observations inherits their intersection. Neither the interval containment nor the recorded timing difference establishes an intrinsic advantage for the energy method. Detailed arithmetic, accounting, and replay instructions remain in [5]. No theorem above uses these numerical intervals.

9 What has been achieved and what remains open

9.1 The present analytic conclusion

The project now has a complete finite-domain identity, a full quadratic calibration, explicit bounds for the energy and both boundary payments after quadratic subtraction, and a source-controlled ordinary-event chart. The source tail estimates in Appendix B make the constants effective without replacing the complete canonical source. The sharper upper bound in Section 6 improves a specified coarse bound on an explicit event-dependent interval. These statements hold with all spatial boundaries, both sheets, the mixed term, and the relative period retained.

The same analysis explains its current limitation. At an ordinary event,

$$\mathcal{E}_\tau = \frac{2\pi}{|A|} \log(1/\tau) + O(1),$$

and the derived upper bounds have this permitted growth. Their constants are finite on each fixed chart but are not a theta-specific obstruction to that chart. The polynomial heat family $z^2 - 2\tau$ satisfies the local mechanism exactly. Consequently ordinary holomorphicity, the heat equation, real symmetry, and a controlled nonvanishing unit cannot alone supply the desired exclusion.

9.2 Endpoint control and coverage

For a collision exclusion through reciprocal energy, one would need a new estimate for the unchanged complete source, for example

$$\sup_{0<\tau\le\tau_0} \mathcal{E}(H(T+\tau,\cdot);D)<\infty \tag{9.1}$$

on a fixed disk containing the collision, with hypotheses not already assuming the conclusion. The logarithmic lower bound would then give a contradiction. No estimate of the form (9.1) is proved here. Alternatively, a source inequality that forces a sufficiently strong upper control of the relative period, or of the full exterior payment after the common constant is fixed, could address the same gap. It must be checked against the complete quadratic calculation.

Even such an ordinary-event estimate would need a coverage argument. The present theorem assumes two real simple roots in a fixed disk and starts from a hypothetical real nondegenerate double event. It does not treat every height, higher multiplicities, arbitrary nonreal pairs, or a moving spatial boundary. An individual simple nonreal zero still has finite reciprocal energy by Proposition 2.2; regular fiber finiteness by itself does not locate zeros on the real axis. Any implication toward RH or toward RH plus simplicity therefore needs an additional global bridge, with its reality regime, endpoint passage, and uniform constants stated separately. The usual canonical heat-flow context is described in [1]; it does not provide the missing energy estimate merely by a change of language.

9.3 Controls on a proposed source-specific advance

A future source-specific inequality should state exactly which canonical properties it uses and test their separate roles. A positive finite weighted-square or finite-mode control may preserve useful positivity while failing the complete normalized reflection identity and the unchanged canonical coefficients. A reflected primitive control may preserve evenness or a transform identity while losing the canonical square-index sequence. An Euler-product modification may change the arithmetic source itself. Such controls answer different questions; none should be said to preserve the entire canonical class merely because one identity survives.

The relevant next analytic task is thus to isolate a complete-source input that controls degeneration beyond the generic chart bounds. The finite identities and paid remainders here specify where that input must enter. A compact-surface or Jensen-based reserve approach would likewise require its own quantitative passage from finite bordered surfaces; the compact bilinear relation alone does not remove the physical boundary or supply the needed uniformity.

A Canonical normalization and source reflection

For completeness we derive the classical normalization used in (1.3). Let

$$\psi(x)=\sum_{n=1}^{\infty} e^{-\pi n^2 x}, \qquad \Theta(x)=1+2\psi(x) \quad (x>0).$$

The Fourier transform of $e^{-\pi x v^2}$, with kernel $e^{-2\pi i v\xi}$, is $x^{-1/2}e^{-\pi\xi^2/x}$. Poisson summation for this rapidly decaying Gaussian therefore gives

$$\Theta(x)=x^{-1/2}\Theta(1/x), \qquad \psi(x)=x^{-1/2}\psi(1/x)+\tfrac{1}{2}(x^{-1/2}-1). \tag{A.1}$$

Set $f(u) = e^u\psi(e^{4u})$ for real u. Direct differentiation of the locally uniformly convergent series gives

$$\frac{f''(u) - f(u)}{8} = 2e^{9u}\psi''(e^{4u}) + 3e^{5u}\psi'(e^{4u}) = \Phi(u). \tag{A.2}$$

The theta identity gives $f(-u) = f(u) + \sinh u$. Applying $\partial_u^2 - 1$, which annihilates $\sinh u$, proves $\Phi(-u) = \Phi(u)$. Differentiating at zero also gives $-f'(0) = f'(0) + 1$, or $f'(0) = -1/2$. These identities concern the complete series.

For $\operatorname{Re} s > 1$, absolute convergence and the gamma integral give

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \int_0^\infty \psi(x)x^{s/2-1}\,dx.$$

Split at $x = 1$, apply (A.1) on $(0, 1)$, and substitute $x \mapsto 1/x$ in its decaying part. The elementary remaining integral is $1/(s-1) - 1/s = 1/(s(s-1))$. Hence

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{s(s-1)} + \int_1^\infty \psi(x)\left(x^{s/2-1} + x^{(1-s)/2-1}\right)\,dx. \tag{A.3}$$

The last integral is entire in s by its exponential decay. Multiplying by $s(s-1)/2$ thus continues ξ as an entire function. Put $v = 2s - 1$ and $x = e^{4u}$ in (A.3). This gives

$$\xi(s) = \frac{1}{2} + (v^2 - 1)\int_0^\infty f(u)\cosh(vu)\,du. \tag{A.4}$$

Twice integrating by parts, using the super-exponential decay of f, f' at $+\infty$ and $\sinh(0) = 0$, yields

$$\begin{aligned}\int_0^\infty (f'' - f)\cosh(vu)\,du &= -f'(0) + (v^2 - 1)\int_0^\infty f(u)\cosh(vu)\,du \\ &= \xi(s).\end{aligned}$$

All these integrals converge for every complex v, so analytic continuation introduces no new restriction. Finally take $s = (1 + \mathrm{i}z)/2$, for which $v = \mathrm{i}z$, and use (A.2). The result is

$$H(0, z) = \int_0^\infty \Phi(u)\cos(zu)\,du = \frac{1}{8}\xi((1 + \mathrm{i}z)/2),$$

as claimed. This derivation also fixes the factor $1/8$ used by the energy and numerical records; a scalar change of H would rescale its reciprocal energy by the inverse scalar modulus.

B Effective complete-source moment tails

The moments used in the event chart retain the entire theta source and the entire integration half-line. This appendix gives explicit upper bounds for replacing them by finitely many modes on a finite interval. For $S \geq 0$, $Y \geq 0$, and an integer $j \geq 0$, recall the notation

$$M_{j,Y}(S) = \int_0^\infty u^j e^{Su^2 + Yu}\Phi(u)\,du. \tag{B.1}$$

Write $\lambda_n = \pi n^2$ and split the source in (1.2) into its complete individual summands,

$$\Phi_n(u) = \lambda_n e^{5u}(2\lambda_n e^{4u} - 3)e^{-\lambda_n e^{4u}}. \tag{B.2}$$

For $u \geq 0$, the factor $2\lambda_n e^{4u} - 3$ is at least $2\pi - 3 > 0$. Thus every Φ_n is positive, and

$$0 < \Phi_n(u) \leq 2\lambda_n^2 e^{9u - \lambda_n e^{4u}}. \tag{B.3}$$

The smaller term in (B.2) is discarded only in this upper estimate; it remains part of the finite source below.

Choose an integer $N \geq 0$ and a real $U \geq 1$. Put $k = N + 1$ and define the exact finite head

$$M_{j,Y}^{N,U}(S) = \sum_{n=1}^{N} \int_0^U u^j e^{Su^2 + Yu} \Phi_n(u)\, du. \tag{B.4}$$

An empty sum is understood as zero. Positivity and Tonelli's theorem, initially allowing infinite values, give the disjoint decomposition

$$\begin{aligned} M_{j,Y}(S) = M_{j,Y}^{N,U}(S) &+ \sum_{n=k}^{\infty} \int_0^{\infty} u^j e^{Su^2 + Yu} \Phi_n(u)\, du \\ &+ \sum_{n=1}^{N} \int_U^{\infty} u^j e^{Su^2 + Yu} \Phi_n(u)\, du. \end{aligned} \tag{B.5}$$

The first omitted piece comprises all omitted modes on the whole half-line. The second comprises only retained modes on their exterior rays. Their index–integration regions do not overlap.

Proposition B.1 (All omitted modes)**.** *Set*

$$\rho_k = \left(\frac{k+1}{k}\right)^4 e^{-\pi(2k+1)}, \qquad b_k = 4\lambda_k - 9 - Y. \tag{B.6}$$

If $b_k > 0$ and $S \leq 8\lambda_k$, then

$$\sum_{n=k}^{\infty} \int_0^{\infty} u^j e^{Su^2 + Yu} \Phi_n(u)\, du \leq L_{j,Y}(S; N) := \frac{2\lambda_k^2 e^{-\lambda_k} j!}{(1 - \rho_k) b_k^{j+1}}. \tag{B.7}$$

In particular $\rho_k < 1$ for every $k \geq 1$.

Proof. For $n \geq k$ and $u \geq 0$, the ratio of consecutive leading terms in (B.3) is

$$\frac{\lambda_{n+1}^2 e^{-\lambda_{n+1} e^{4u}}}{\lambda_n^2 e^{-\lambda_n e^{4u}}} = \left(\frac{n+1}{n}\right)^4 e^{-\pi(2n+1)e^{4u}} \leq \rho_k.$$

Both the rational factor and the exponential factor are bounded above by their values at $n = k$, $u = 0$. Also $\rho_k \leq 16 e^{-3\pi} < 1$. Geometric summation therefore gives

$$\sum_{n=k}^{\infty} \Phi_n(u) \leq \frac{2\lambda_k^2}{1 - \rho_k} e^{9u - \lambda_k e^{4u}}. \tag{B.8}$$

For $u \geq 0$, Taylor's inequality for the positive exponential series is

$$e^{4u} \geq 1 + 4u + 8u^2. \tag{B.9}$$

Consequently

$$\begin{aligned} Su^2 + (9 + Y)u - \lambda_k e^{4u} &\leq -\lambda_k - (4\lambda_k - 9 - Y)u + (S - 8\lambda_k)u^2 \\ &\leq -\lambda_k - b_k u. \end{aligned}$$

Multiply (B.8) by $u^j e^{Su^2+Yu}$ and integrate. The elementary integral

$$\int_0^\infty u^j e^{-bu}\,du = \frac{j!}{b^{j+1}} \qquad (b>0)$$

follows by scaling and j integrations by parts. It gives exactly (B.7). □

Proposition B.2 (Retained exterior rays)**.** *For each retained index $1 \le n \le N$, suppose that*

$$b_n(U) = 4\lambda_n e^{4U} - 9 - 2SU - Y > 0, \qquad S \le 8\lambda_n e^{4U}. \tag{B.10}$$

Then

$$\int_U^\infty u^j e^{Su^2+Yu}\Phi_n(u)\,du \le R_{n,j,Y}(S;U), \tag{B.11}$$

$$R_{n,j,Y}(S;U) := 2\lambda_n^2 e^{SU^2+(9+Y)U-\lambda_n e^{4U}} \sum_{h=0}^{j} \binom{j}{h} \frac{U^{j-h}h!}{b_n(U)^{h+1}}. \tag{B.12}$$

Proof. Set $u = U + v$, where $v \ge 0$. Applying (B.9) to v gives

$$\begin{aligned} S(U+v)^2 + (9+Y)(U+v) - \lambda_n e^{4U}e^{4v} \le{}& SU^2 + (9+Y)U - \lambda_n e^{4U} \\ & - b_n(U)v + (S - 8\lambda_n e^{4U})v^2. \end{aligned}$$

The quadratic coefficient is nonpositive by hypothesis. Thus (B.3) bounds the exterior integral by

$$2\lambda_n^2 e^{SU^2+(9+Y)U-\lambda_n e^{4U}} \int_0^\infty (U+v)^j e^{-b_n(U)v}\,dv.$$

The binomial theorem and the same elementary exponential integrals give

$$\int_0^\infty (U+v)^j e^{-bv}\,dv = \sum_{h=0}^{j} \binom{j}{h} \frac{U^{j-h}h!}{b^{h+1}},$$

which proves (B.11)–(B.12). □

Corollary B.3 (Complete finite enclosure)**.** *Under the hypotheses of Propositions B.1 and B.2,*

$$M_{j,Y}^{N,U}(S) \le M_{j,Y}(S) \le M_{j,Y}^{N,U}(S) + L_{j,Y}(S;N) + \sum_{n=1}^{N} R_{n,j,Y}(S;U). \tag{B.13}$$

For each fixed S, Y, j, admissible N, U exist and the sum of the displayed tail bounds can be made arbitrarily small.

Proof. Insert the two upper bounds into the positive disjoint identity (B.5). To verify the final assertion, first increase N. Then $b_k = 4\pi k^2 - 9 - Y$ is positive, $8\pi k^2 \ge S$, $\rho_k \to 0$, and the right-hand side of (B.7) tends to zero because of $e^{-\pi k^2}$. With such an N fixed, increase U. For the finitely many retained indices, the term e^{4U} dominates both U and the constant S; hence (B.10) eventually holds simultaneously. In (B.12), the negative exponent $-\lambda_n e^{4U}$ dominates $SU^2 + (9+Y)U$, while the finite sum has only polynomial powers of U and inverse positive powers of $b_n(U)$. Thus every $R_{n,j,Y}(S;U)$ tends to zero. This proves the assertion and also proves finiteness of (B.1) directly. □

These bounds estimate the complete positive moments. They do not impose separate reflection identities on a source head or tail, and they make no replacement of the canonical coefficient sequence. In particular, the finite head in (B.4) retains both terms of every summand in (B.2).

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[5] E. Baker research project, *Actual regular theta fiber comparison*, Note 04, certificate `03_ACTUAL_THETA_COMPARISON_20261022.json`, and Review 03, 22 October 2026. Prepared for Edward Baker with substantial LLM assistance, `gpt-6.1-sol`, effort ultra. Accompanying repository directory `papers/investigations-4/09_reciprocal_hodge_energy/`. The certificate and its three replay scripts are in `numerics/`; the implementation-specific comparison and all error budgets are documented in Note 04 and the fixed comparison contract.